MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSCCMA200 – Data Analytics Using R

Duration:1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each Course Outcome (CO) to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

(Answer all questions, each question carries 2 marks.)

1. Which function is used to list variables in **R**? [U] [CO1]
2. How will you create a matrix in **R**? [U] [CO1]
3. What is the purpose of edit() function in **R**? [U] [CO2]
4. What is the use of melt() function in **R**? [U] [CO2]
5. How will you calculate frequencies of a vector in **R**? [U] [CO3]

(5X2=10 Marks)

Part B

Short Answer Questions

(Answer 5 out of 7 questions, each question carries 5 marks.)

6. What is **R** and why is it widely used in data analytics? [U] [CO1]
7. Discuss relational operators in **R**. [U] [CO1]
8. What is the difference between **break** and **next** statements in **R** loops? [U] [CO1]
9. What is a data frame in **R**? [U] [CO2]
10. Differentiate between rbind() and cbind(). [U] [CO2]
11. Define correlation and state its significance in data analysis. [An] [CO3]
12. How is mean calculated in **R** with the trim option? [An] [CO3]

(5X5=25 Marks)

Part C

Essay Questions

(Answer 1 out of 2 questions, each question carries 15 marks.)

13. Explain the creation of data frames, methods of accessing data, and the use of functions such as dim(), str(), summary(), head(), and tail() for data analysis. [U] [CO2]
14. Describe the data visualization techniques available in **R**. [An] [CO3]

(1X15=15 Marks)